

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. (ME) Examination

May 2014

ME104 Basics of Mechanical Engineering

DATE: 20.05.2014, TUESDAY TIME: 01:30 P.M. TO 04:30 P.M. MAXIMUM MARKS: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Property table is given at the end of paper.

SECTION-I

Q-1 (a) Explain construction and working of Cochran boiler with neat sketch. [07]

(b) Differentiate between clutch and brake. [04]

OR

(b) Explain internal expanding shoe brake with neat sketch. [04]

Q-2 (a) Explain steam formation with T-h diagram in detail. [07]

(b) A boiler generates 7.5 kg of steam/ kg of coal burnt at a pressure of 11 bar. The feed water temperature is 70°C. Boiler efficiency is 75%. Factor of evaporation = 1.15. take $C_{ps} = 2.1 \text{ kJ/kg K}$. Calculate (i) Degree of Superheat (ii) Temp of steam generated (iii) calorific value of coal (iv) Equivalent evaporation in kg of steam/ kg of coal. [05]

Pressure (bar)	$T_{\text{sat}} (^{\circ}\text{C})$	Specific enthalpy in kJ/kg		
		h_f	h_{fg}	h_g
11	184.07	781.12	1998.5	2779.7

OR

Q-2 (a) Give classification of Mechanical drive and explain helical gear and rack and pinion gear. [07]

(b) Explain construction and working of centrifugal clutch with neat sketch. [05]

- Q-3 (a)** Explain construction and working of fusible plug with neat sketch. [05]
- (b)** What is accessory in boiler? Explain construction and working of air pre-heater with neat sketch. [07]

OR

- Q-3 (a)** Explain various type of belt drive in detail with neat sketch. [07]
- (b)** Give comparison of belt drive, chain drive, and gear drive. [05]

SECTION – II

- Q- 4(a)** Define: pressure, heat, specific heat, enthalpy, power. [05]
- (b)** Give statement of first law of Thermodynamic and derive $PV = mRT$ for perfect gas. [05]
- Q-5 (a)** Explain working of 4 stroke petrol engine with neat sketch. [07]
- (b)** Following data refers to 2 stroke SI engine. [07]

Torque = 640 Nm, Crank Speed = 350 rpm, Mean effective Pressure = 5.6 bar, Bore=21 cm, stroke = 28 cm, fuel consumption = 8.16 kg/hr, calorific value = 42705 kJ/kg.

Calculate: (i) mechanical efficiency (ii) indicated thermal efficiency (iii) brake thermal efficiency (iv) SFC (v) relative efficiency if compression ratio is 11.

OR

- Q-5 (a)** Explain working of Two stroke Diesel engine with neat sketch. [07]
- (b)** Give constructional detail of internal combustion engine with neat sketch. [07]
- Q-6(a)** Define refrigeration and explain working of vapour compression refrigeration system with flow diagram. [06]
- (b)** What is air conditioning? Explain window air conditioning system in detail with sketch. [05]

OR

- Q-6 (a)** Explain vapour absorption system in detail with proper flow diagram. [06]
- (b)** What is refrigerant? Explain required properties of refrigerant, [05]
